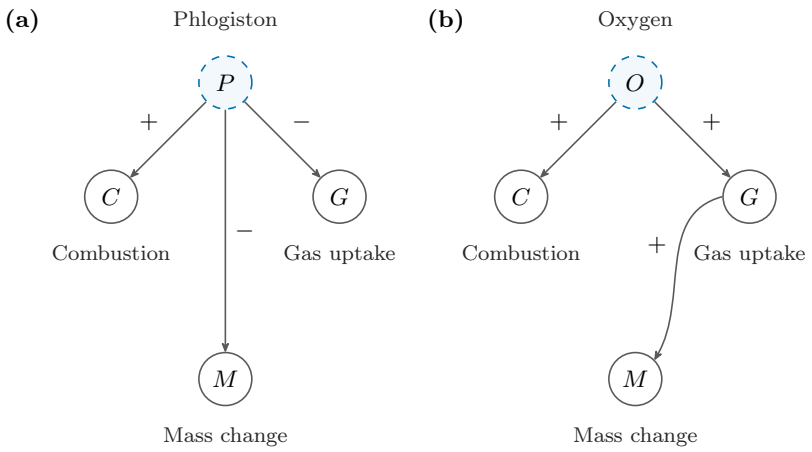
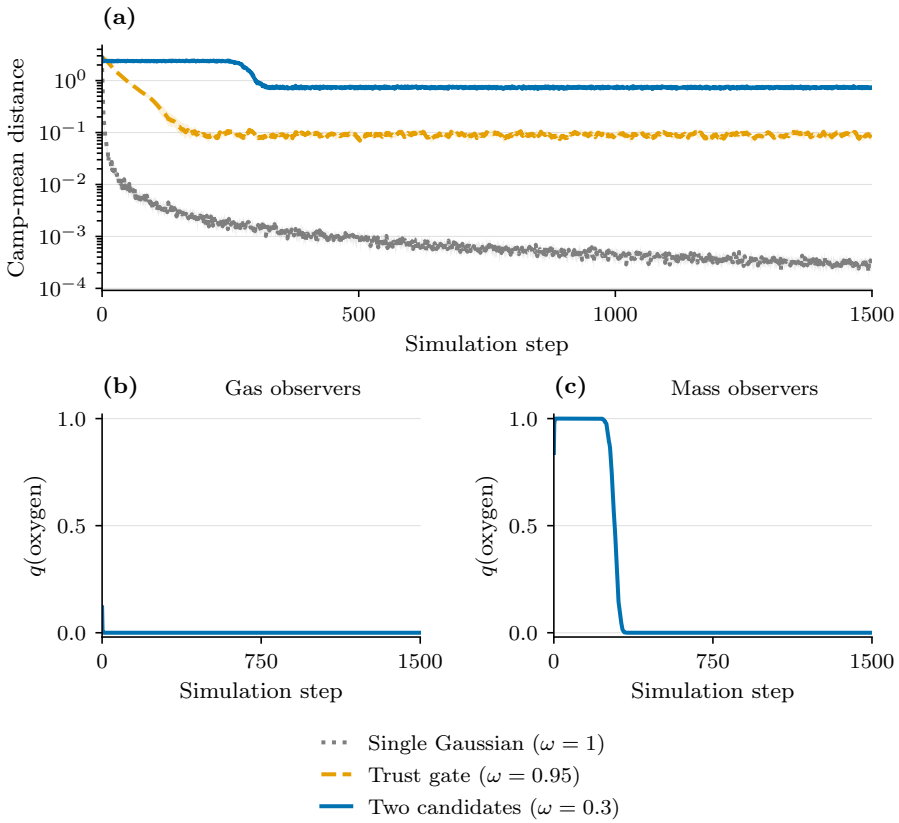
Four-variable experiment: figure 1



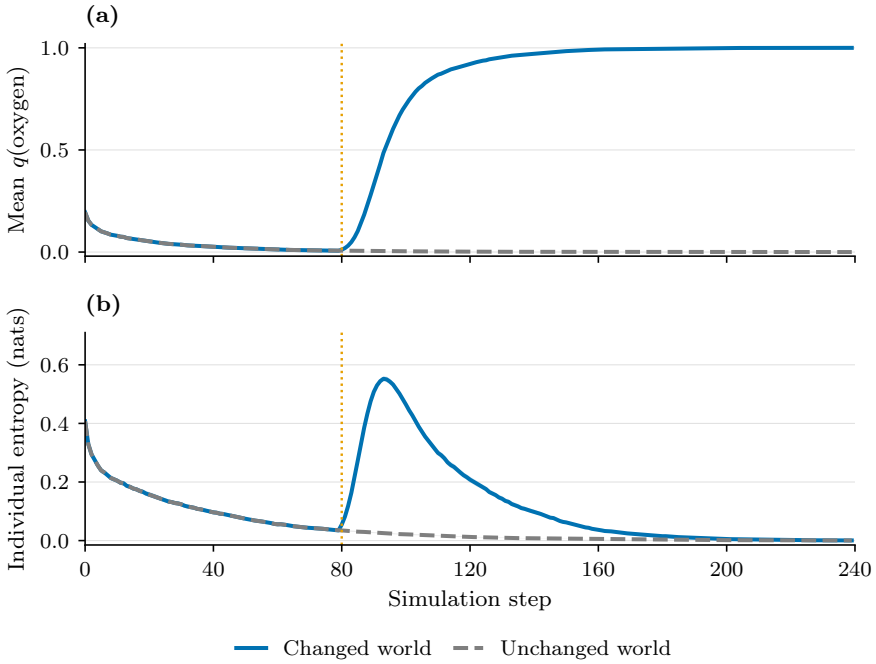
Two supplied prior theories organize the same three observable commitments. The hidden explanatory variable is interpreted as phlogiston release P in (a), or oxygen participation O in (b). Observable commitments are combustion C, gas uptake G and mass change M. The phlogiston candidate links its core directly to all three; the oxygen candidate links its core to C and G, and G to M. Arrows are linear-Gaussian belief dependencies, not a historically calibrated chemical model. Signs show coefficient directions; magnitudes are 0.8 at the core and 0.9 for G to M. Dashed nodes are unobserved. The same array coordinate holds the candidate-specific hidden variable; it is not shared evidence that P and O denote one physical quantity. Single-Gaussian controls operate on the three shared observable commitments. No aggregate of background commitments remains.

Four-variable experiment: figure 3



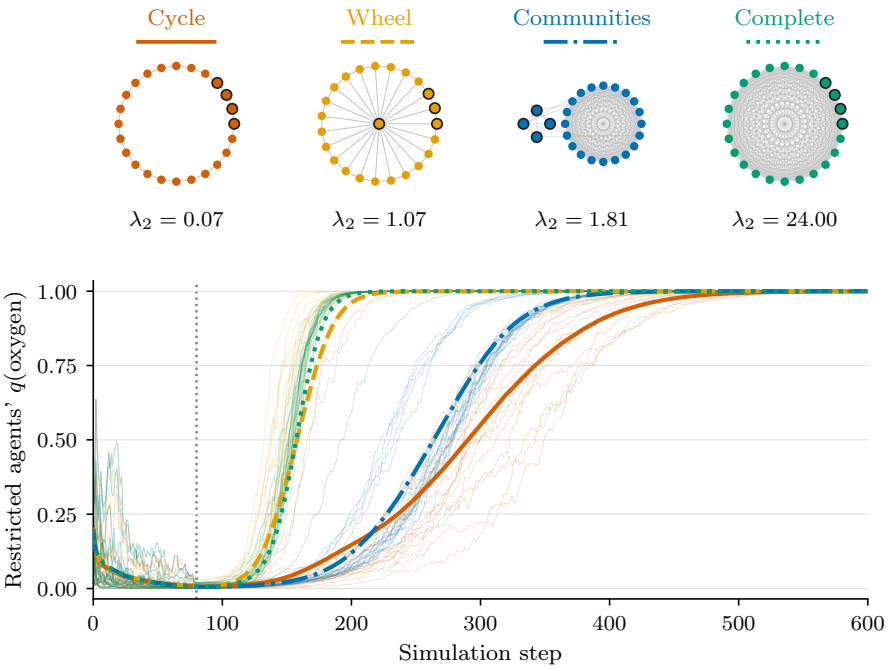
A persistent parameter gap does not preserve opposing theories in this four-variable world. Two camps of eight agents attend G or M in the fixed mixed world (C,G,M)=(1,-1,1) and communicate on a complete graph. The gate downweights social influence from agents with differing observable means; it is not a sensory-reliability update. (a) Euclidean distance between camp-mean commitments on C, G and M; the two-candidate condition reads out each agent's highest-probability candidate. The logarithmic axis displays all positive values without clipping. (b,c) The camps' oxygen probabilities in that candidate condition, allowing a parameter gap to be distinguished from sustained opposing model choices. The inherited comparison uses memory 1 for the plain Gaussian, 0.95 for the gate and 0.3 for the two candidates. Anchors also differ, so this comparison does not isolate representation alone. Lines average 20 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure 4



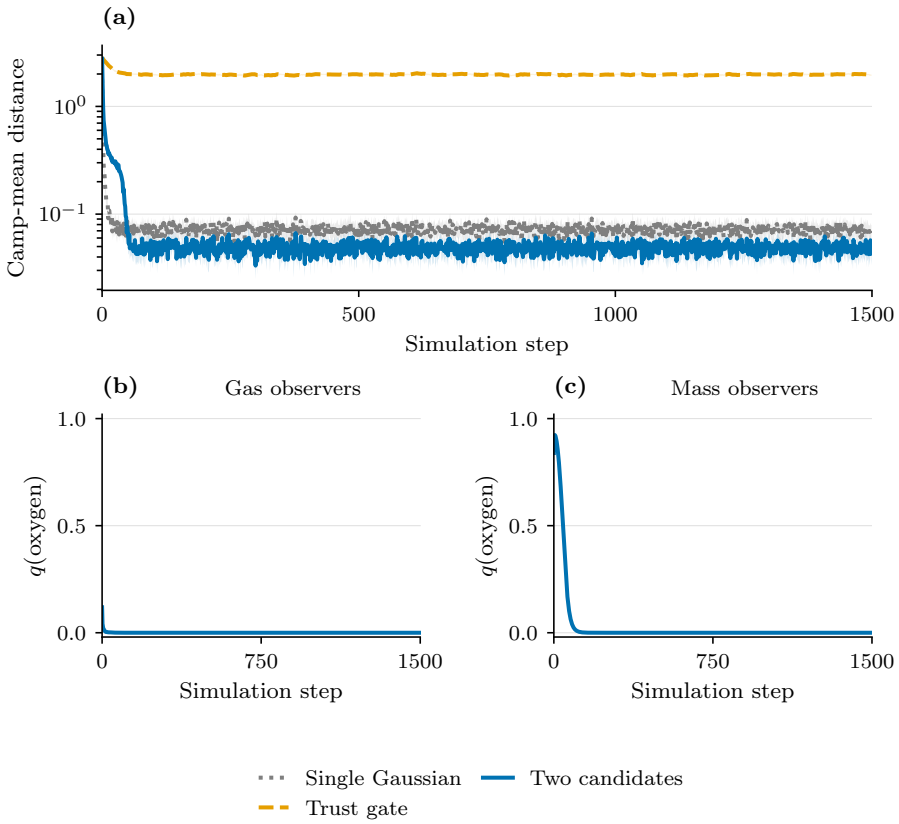
The cycle experiment tests switching between two supplied theories. Sixteen agents on a complete graph start with equal candidate probabilities. The observable world changes at step 80 (dotted line): C stays at +1 while G and M switch from -1 to +1. The grey control does not change. (a) Probability assigned to oxygen, averaged over agents within each seed. (b) Individual posterior entropy, computed before averaging agents and seeds. Parameter memory is 0.9, model-score sharing 0.02 and observation noise standard deviation 1. Both candidates remain available throughout; no theory is generated. The first plotted point follows the first update. Lines average 50 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure 9



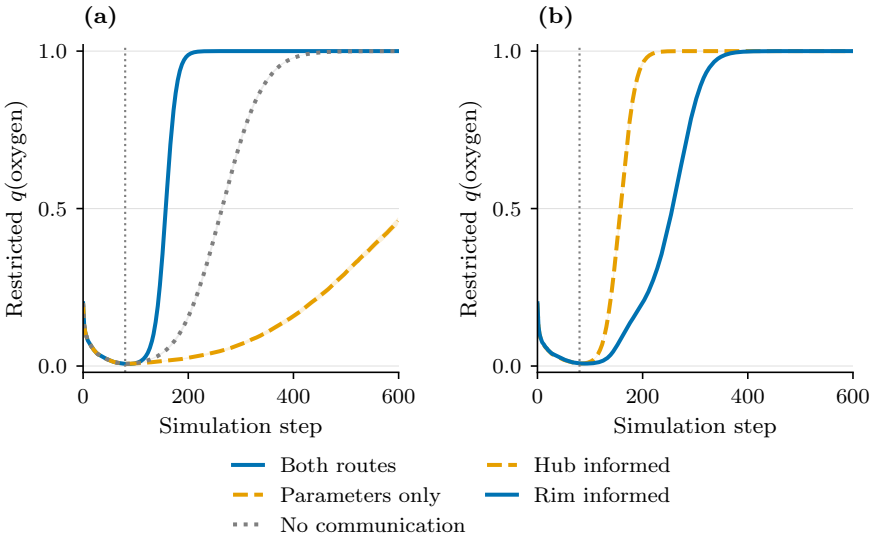
Contact topology changes how the community responds to the same evidence. All 24 nodes and edges are shown; black outlines mark four full-attention agents. After the world changes at step 80, the other 20 agents assign weight 0.01 to gas-uptake and mass-change observations; combustion remains fully attended. Thin lines show every restricted agent from seed 0; heavy lines average restricted agents within each seed and then average seeds. The graph definitions, informed positions and row-normalized self-plus-neighbour weights match the accepted experiment. Memory is 0.9 and model-score sharing 0.02. Lambda2 is the unnormalized graph Laplacian's second eigenvalue. Density and source placement vary alongside topology; this is not an isolated test of that eigenvalue. Lines average 50 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S1



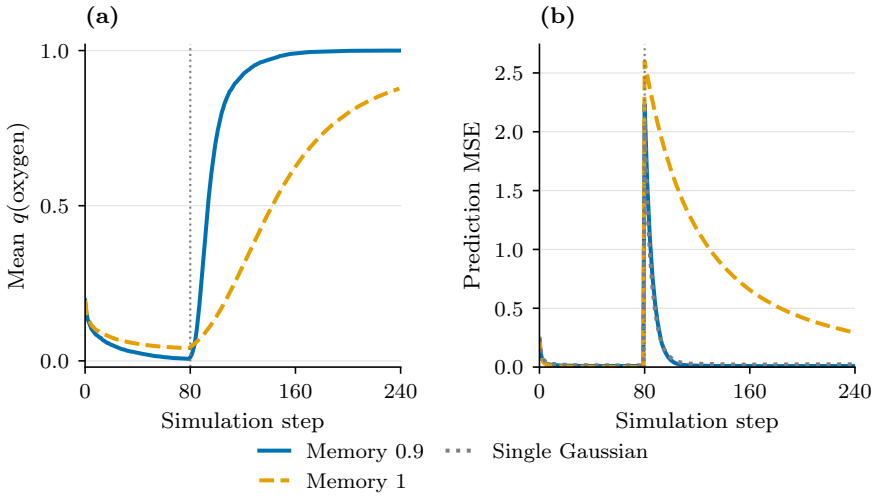
A persistent parameter gap does not preserve opposing theories in this four-variable world. Two camps of eight agents attend G or M in the fixed mixed world (C,G,M)=(1,-1,1) and communicate on a complete graph. The gate downweights social influence from agents with differing observable means; it is not a sensory-reliability update. (a) Euclidean distance between camp-mean commitments on C, G and M; the two-candidate condition reads out each agent's highest-probability candidate. The logarithmic axis displays all positive values without clipping. (b,c) The camps' oxygen probabilities in that candidate condition, allowing a parameter gap to be distinguished from sustained opposing model choices. All three conditions use memory 0.9, but the gated condition has camp-specific anchors and the candidate condition has candidate-specific anchors; this is not a pure intervention on representation alone. Lines average 20 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S2



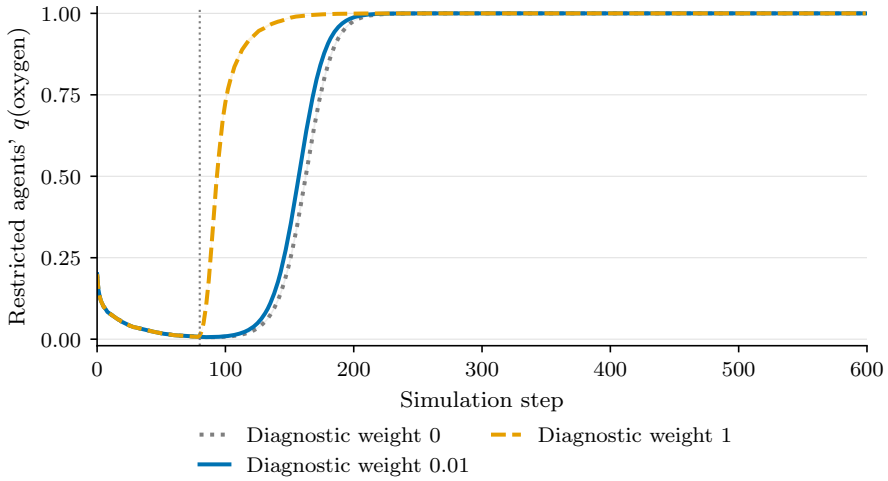
Communication routes and informed-agent placement are separate interventions. (a) On the complete graph, both information-sharing routes operate, only parameters are shared, or all social sharing is disabled. Restricted agents still receive gas and mass evidence at weight 0.01. (b) The wheel's full-attention agents include the hub or occupy four rim nodes. Each curve averages the other 20 agents within each seed. In (a), all conditions branch at step 80 from exactly the same pre-change state; the communication intervention and attention restriction start together. Lines average 50 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S3



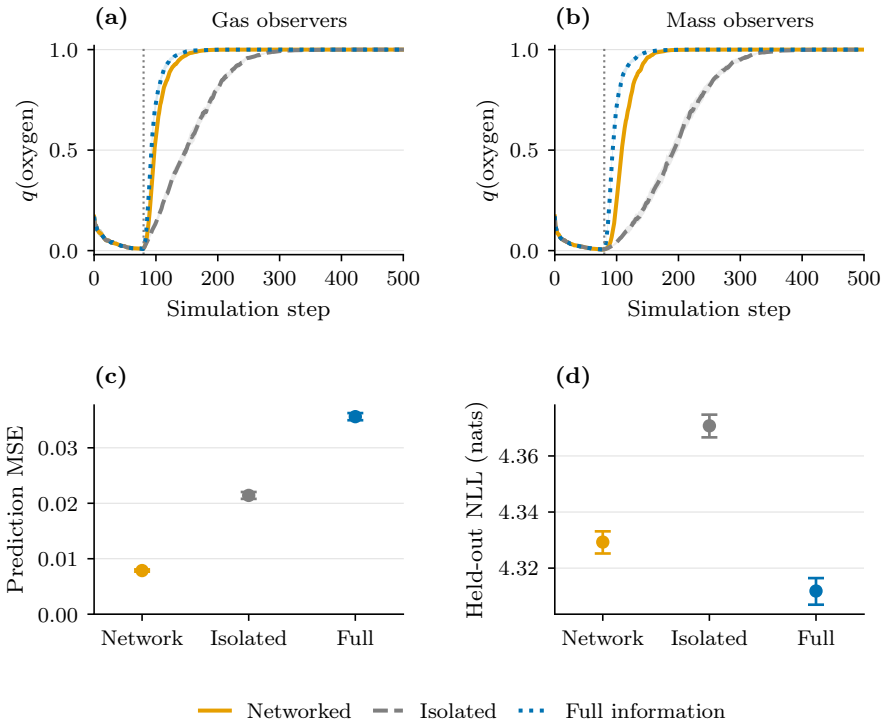
A memory control tests whether adaptation depends on forgetting. The world changes at step 80. (a) Mean candidate probability with parameter memory 0.9 or 1, retaining the same priors, observations and communication settings. (b) Noiseless observable prediction MSE adds a single-Gaussian comparator, formed by information pooling of the candidates' observable marginal priors, with memory 0.9. This comparator has no candidate probability. A finite run identifies delay or failure to switch within its horizon; it cannot establish permanent lock-in. Lines average 50 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S4



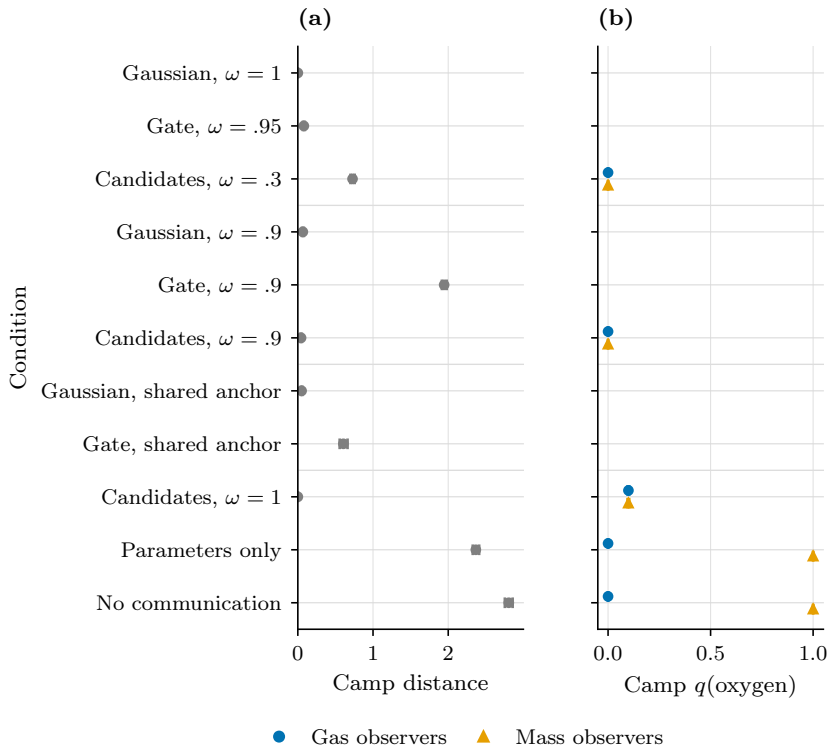
A direct-evidence control separates reduced attention from no diagnostic observations. Twenty agents on a complete graph assign the indicated weights to gas and mass observations after step 80; four source agents always assign weight 1. All conditions retain full combustion attention and both communication routes. They share the same pre-change state and private noise streams. Weight zero removes the restricted agents' diagnostic input but not information arriving through neighbours. Lines average 50 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S5



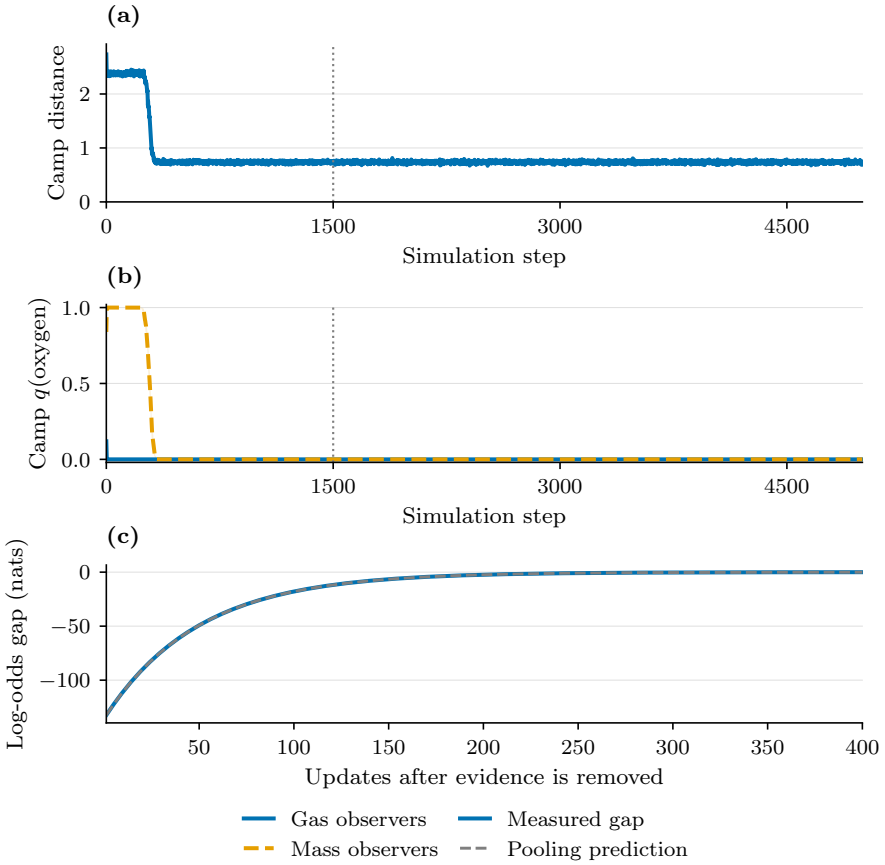
Social sharing accelerates correction, while full information gives the best predictive density. At step 80 the world changes to the oxygen reference pattern. Agents branch from one shared pre-change state into networked selective observation, isolated selective observation, or a reference with both diagnostic channels available individually. (a,b) Oxygen probabilities for the gas- and mass-observing camps during the first 500 of 1,500 updates; reference agents in both plotted groups observe both channels. Points in (c,d) average the last 200 updates within each seed; whiskers use the same whole-seed bootstrap as the ribbons. (c) Mean squared error of the posterior mixture mean on C,G,M against the noiseless world vector, averaged over coordinates and agents within each seed. Sensor noise variance is excluded from this error; it is not a held-out log-likelihood. (d) Negative log density of the full posterior predictive mixture on four independent withheld observations per agent and update, in nats per three-coordinate observation. Lower is better in (c,d); the NLL axis is expanded to show differences. The reference is not a centralized observer pooling all agents' observations. These data test adaptation to a prescribed mean pattern, not recovery of the true causal wiring. Lines average 20 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.

Four-variable experiment: figure S6



The full disagreement-control matrix reports parameter and candidate outcomes separately. Each row summarizes step 1499: (a) distance between camp-mean MAP observable commitments; (b) each camp's mean oxygen probability where candidates exist. Empty probability rows are single-Gaussian conditions, not missing runs. Memory is shown explicitly; the shared-anchor Gaussian controls use memory 0.9, while communication controls use candidate memory 0.3. Other Gaussian controls retain camp-specific anchors. Points average 20 seeds; bars are pointwise 95% percentile-bootstrap intervals (2,000 whole-seed resamples).

Four-variable experiment: figure S7



A longer run separates a distance plateau from persistent competing model choices. The candidate condition with memory 0.3 continues to 5,000 updates without changing its mixed world or selective evidence; the dotted line marks the saved continuation boundary, not an intervention. (a) MAP commitment distance. (b) Candidate probabilities in the two camps. (c) A separate branch from update 1499 removes all incoming evidence and sets memory to 1. The signed camp log-odds gap follows the pure-pooling recurrence gap(t) = gap(0) times (1 - alpha_m)^t. This tests the source of disagreement, not adaptation to a new world. Lines average 20 seeds; ribbons are pointwise 95% percentile-bootstrap intervals for the seed mean (2,000 resamples of complete seed curves). No smoothing or interval clipping is used.